

Week 5: Are These Airbnb Differences Real?

T-tests in SPSS and PSPP (practice handout)

NCKU | Maxwell K. Hsu

What you will learn

Last week you compared two categories with a Chi-square test. This week the outcome is a **number** (a price or a score), and you ask: is a difference between averages **real**, or could it be **chance**? A **t-test** (t 檢定) helps you answer that question.

1. Turn a business question into a research question and two hypotheses.
2. Decide when a t-test fits a question, and which kind of t-test to use.
3. Run an independent-samples, a paired, and a one-sample t-test in SPSS or PSPP.
4. Read the output, and report a p-value, an effect size, and a confidence interval in plain English.

Why this matters to you: managers often ask, "Is this difference real?" A t-test gives an honest answer. Knowing how to explain it makes your analysis trustworthy.

How this handout works

- Parts 1-5 are the **core track**. Everyone does them.
- Part 6, **Going further**, is optional and not graded.
- Each part follows the same steps: research question, why a t-test fits, hypotheses, look first, run the test, read the output, report it. If you feel lost, read the **Stuck?** tip at the end of the part.
- At the end, **Check your results** lists the numbers you should get for Parts 1-4.

Which tool should I use?

You may do this exercise in **Google Colab**, **IBM SPSS**, or **GNU PSPP**. All three use the same data and the same questions, and they give the same answers (n, means, mean differences, t, df, p-values, and 95% confidence intervals). Pick the tool you will use in your own research.

Tool	Cost	How you run it	Cohen's d
Google Colab	Free (Google account)	The notebook 03_ttest_airbnb.ipynb	Printed by the notebook
IBM SPSS (version 27 or later)	University licence	Menus, or the syntax file	Effect Sizes table

Tool	Cost	How you run it	Cohen's d
GNU PSPP	Free: gnu.org/software/pspp	Week5_Ttest.sps Menus (similar to SPSS), or the same syntax file	Compute it by hand (Part 1 shows how)

This handout shows the SPSS menus first, then the **syntax** (the commands behind the menus). The syntax works in both SPSS and PSPP; it was checked with GNU PSPP 1.4.1.

To run the syntax file: put `Week5_Ttest.sps` and `nyc_airbnb_ttest.sav` in the same folder. Open the syntax with File > Open > Syntax. Select the commands for one part and click Run > Selection (or run everything with Run > All). Run the parts **in order**: Part 6 uses `ln_price`, which Part 1 creates.

Open the data

Click File > Open > Data and choose `nyc_airbnb_ttest.sav`. In syntax:

```
* Open the data file (30,259 NYC listings).
GET FILE='nyc_airbnb_ttest.sav'.
```

If GET FILE fails (SPSS looks in its working folder, not in the folder of the syntax file), open the data with File > Open > Data, then run the syntax from the next command.

Check: the Data View should show 30,259 cases (rows), one per NYC Airbnb listing. Nothing has been removed.

Variable View tour. Click the Variable View tab at the bottom of the window. Each row is one variable:

Variable	What it holds
<code>id</code>	Listing ID
<code>host_type</code>	1 = Superhost, 2 = Other host
<code>room_type</code>	1 = Entire home/apt, 2 = Private room, 3 = Hotel room, 4 = Shared room
<code>accommodates</code>	Number of guests
<code>price</code>	Nightly price (USD)
<code>price_outlier</code>	0 = No, 1 = Yes (extreme price, flagged in the Week 3 cleaning)
<code>review_scores_rating</code>	Overall rating (0-5)
<code>review_scores_cleanliness</code>	Cleanliness rating (0-5)
<code>review_scores_location</code>	Location rating (0-5)

Look at the **Values** column (the number codes and their labels) and the **Measure** column (nominal for groups, scale for numbers). 349 listings have an unknown host type. They are **system-missing** (a dot in the Data View). Some listings also have no price or no rating. SPSS and PSPP leave missing values out of each test automatically, so the n in each output table is smaller than 30,259.

Stats in 5 minutes

Read this once. Come back to it whenever you need it.

Term	Plain English	Airbnb example
Mean (平均數)	The average. Add all values and divide by how many there are.	The average nightly price.
Median (中位數)	The middle value when you sort from low to high. Extreme values do not move it much.	Half of listings cost less than the median price.
Sample (樣本)	The data we have. We use it to learn about a bigger group.	These listings stand for NYC Airbnb listings in general.
Hypothesis (假設)	A claim we test. H0 (null, 虛無假設) says "no difference". H1 (alternative, 對立假設) says "there is a difference".	H0: superhosts and other hosts charge the same on average.
p-value (p 值)	If there were really no difference, how surprising would our result be? A small p (below .05) means "probably a real difference".	p = .30: not surprising, no evidence of a difference. p < .001: very surprising, a real difference is likely.
Effect size, Cohen's d (效果量)	How big the difference is, in standard units. Read it like a traffic light: under 0.2 tiny, 0.2 to 0.5 small, 0.5 to 0.8 medium, 0.8 or more large.	d = 0.05: the groups are almost the same, even if p is small.
95% confidence interval (95% 信賴區間)	The likely range for the true difference.	"Between 2 and 9 dollars more per night."

Two rules to remember

1. A small p-value says a difference is probably **real**. It does not say the difference is **big**.
2. Always report the effect size next to the p-value.

Choosing a test

Start from your question, not from the test.

Ask yourself	If the answer is...	Use	SPSS menu
Is my outcome a number (price, score)?	Yes	A t-test can fit. (If it is a category, such as yes/no, use a Chi-square test, as in Week 4.)	
How many groups do I compare?	Two separate groups (each listing is in one group only)	Independent-samples t-test (獨立樣本 t 檢定) , Parts 1, 2, 5	Analyze > Compare Means > Independent-Samples T Test
	Two scores from the same listings	Paired-samples t-test (成對樣本 t 檢定) , Part 3	Analyze > Compare Means > Paired-Samples T Test
	One group against a fixed target number	One-sample t-test (單一樣本 t 檢定) , Part 4	Analyze > Compare Means > One-Sample T Test
Do a few extreme values dominate?	Yes	Check the medians and rerun without the outliers (Part 2), or use a rank-based test (Part 6)	Data > Select Cases; Analyze > Nonparametric Tests > Legacy Dialogs

In SPSS 29 and later, the menu is called Analyze > **Compare Means and Proportions**.

Reading SPSS output

SPSS and PSPP print the same kinds of tables. Five things to look for:

1. **Sig. (2-tailed)** is the **p-value**. It is two-tailed: it tests for a difference in either direction.
2. **".000" means "p < .001"**. It does not mean that p is zero: SPSS rounds to three decimals, and a p-value is never exactly zero. Write **p < .001** in your report.
3. **Levene's Test for Equality of Variances (變異數同質性檢定)** appears only in the independent-samples table. It decides between two rows: "Equal variances assumed" (Student's t-test) and **"Equal variances not assumed"** (Welch's t-test). In this course we read the Welch row, the safe default (Part 1 explains why).
4. **Mean Difference** is the first group's mean minus the second group's mean (for example, superhosts minus other hosts). A negative value means the first group is lower.

5. **95% Confidence Interval of the Difference** (Lower, Upper) is the likely range for the true difference. If it does not include 0, the difference is significant at .05.

SPSS also drops the zero before the decimal point (".94" means 0.94). In your report, write the zero for numbers that can be larger than 1 (means, differences, d), and leave it out for p-values ($p = .036$).

SPSS users

The numbers in this handout are what PSPP prints. SPSS often prints more decimals (for example, $t = -2.099$ instead of -2.10). Round your SPSS numbers to the decimals shown in this handout before you compare them. In SPSS 27 or later, the p-value column is called **Significance**, with two columns: **One-Sided p** and **Two-Sided p**. Always read **Two-Sided p** (never One-Sided p); it is the same p-value as "Sig. (2-tailed)".

Part 1: Do superhosts charge a different price?

You are here: Part 1 of 5 (core).

Research question

Airbnb gives the **superhost** (超讚房東) badge to experienced, highly rated hosts. Pricing question: **Do superhosts charge a different nightly price from other hosts?** If they do, the badge may support a price premium.

Why a t-test fits

- The outcome is a **number**: nightly price.
- There are **two separate groups**: each host is either a superhost or not.
- We compare the **averages** of the two groups.

So we use an **independent-samples t-test**. We use **Welch's version**, the safe default, because the two groups have different sizes and spreads.

Hypotheses

- **H0**: Superhosts and other hosts charge the same average price.
- **H1**: Superhosts and other hosts charge different average prices.

Look first

Always look at the data before you test it. A few listings cost thousands of dollars a night, so a boxplot of raw prices is squashed flat. Make a log price first, then draw one box per group.

- **SPSS menus:** Transform > Compute Variable. Target Variable: ln_price; Numeric Expression: LN(price); OK. Then Analyze > Descriptive Statistics > Explore. Dependent List: ln_price; Factor List: host_type; Display: Plots; Plots: Boxplots, Factor levels together; OK.

Check: the line inside each box is the median. Are the two medians far apart?

PSPP users

PSPP's Explore (EXAMINE) also prints an empty "Tests of Normality" table. Ignore it; the boxplot is what you need.

Run the test

- **SPSS menus:** Analyze > Compare Means > Independent-Samples T Test. Test Variable(s): price. Grouping Variable: host_type; click Define Groups, type 1 for Group 1 and 2 for Group 2; Continue; OK. In SPSS 27 or later, keep **Estimate effect sizes** ticked.
- **Syntax** (Part 1 of Week5_Ttest.sps):

* Make a log price so very high prices do not squash the boxplot.

```
COMPUTE ln_price = LN(price).
```

* Look first: boxplots of log price for each host type.

```
EXAMINE VARIABLES=ln_price BY host_type /PLOT=BOXPLOT /STATISTICS=NONE.
```

* Run the t-test of price for superhosts (1) versus other hosts (2).

```
T-TEST GROUPS=host_type(1 2) /VARIABLES=price.
```

Read the output

The **Group Statistics** table shows 6549 superhosts with a mean price of 267.63 and 14688 other hosts with a mean of 281.81.

The **Independent Samples Test** table has two rows:

- **Levene's test** asks whether the two groups have the same spread (variance). Here $F = 6.45$ and $\text{Sig.} = .011$, below .05: the spreads differ, which is one more reason to read the Welch row.
- Read the row **"Equal variances not assumed"** (Welch's t-test). Welch's test does not need equal spreads, so it is safe whether or not Levene's test is significant. This matches the Colab notebook.

In the Welch row: $t = -2.10$, $df = 18708.20$, $\text{Sig. (2-tailed)} = .036$, Mean Difference = -14.18, and the 95% Confidence Interval of the Difference runs from -27.43 to -.94. The decimal degrees of freedom are normal for Welch's test.

Effect size. In SPSS 27 or later, the **Independent Samples Effect Sizes** table shows Cohen's d (use the "Cohen's d" row). PSPP does not print this table; compute d by hand as shown in the box below.

! Why your Cohen's d may differ slightly

Cohen's d divides the mean difference by a standard deviation (SD). The three tools do not all use the same SD for two separate groups:

Tool	Where d comes from	Formula for two separate groups
SPSS 27 or later	Independent Samples Effect Sizes table (also shows Hedges' g and Glass's delta)	Mean difference / pooled SD
PSPP	No effect-size table; compute it by hand from the Group Statistics table	Mean difference / pooled SD (formula below); the rounded table values add small rounding differences
Colab notebook	independent_test helper	Mean difference / $\sqrt{((\text{var_a} + \text{var_b}) / 2)}$

Pooled SD formula (for PSPP users):

$$\text{pooled SD} = \sqrt{\frac{(n_1 - 1) SD_1^2 + (n_2 - 1) SD_2^2}{n_1 + n_2 - 2}} \quad d = \frac{\text{Mean Difference}}{\text{pooled SD}}$$

Worked example (Part 1, PSPP's Group Statistics table): pooled SD = $\sqrt{((6,548 \times 378.07^2 + 14,687 \times 591.59^2) / 21,235)} = 534.92$; $d = -14.18 / 534.92 = -0.027$. The Colab notebook prints $d = -0.029$ because it averages the two variances without weighting by group size. Both round to -0.03, and both are tiny.

- Differences in the second or third decimal are expected.
- The size word (tiny, small, medium, large) should not change.
- Report which tool and which formula you used, for example "Cohen's d (pooled SD) = -0.03".
- SPSS also shows **Hedges' g**, a small-sample correction. At these sample sizes it is almost the same as d.
- For the paired and one-sample tests (Parts 3 and 4), all three tools use the same formula (mean difference / SD of the differences; difference from the target / SD), so only rounding differs. Rounding matters most when the numbers are small: PSPP's Part 3 table shows a mean difference of -.07 and an SD of .44, which gives -0.159, while the unrounded value is -0.149. The size word is the same.

How to report it

Superhosts (M = 267.63, SD = 378.07, n = 6,549) charged USD 14.18 less per night than other hosts (M = 281.81, SD = 591.59, n = 14,688) on average. Welch's t-test found a statistically significant difference, $t(18708.20) = -2.10$, $p = .036$, 95% CI [-27.43, -0.94]. But the effect was tiny (Cohen's $d = -0.03$).

Stuck? Reread "Stats in 5 minutes". Or ask an AI tool: "Explain this result to a first-year student in three sentences:" and paste the two output tables.

Part 2: Is that difference real, or caused by a few extreme listings?

You are here: Part 2 of 5 (core).

Research question

Is the price difference in Part 1 real, or is it caused by a few listings with extreme prices?

In the Week 3 cleaning notebook we flagged price **outliers** (離群値) in the variable `price_outlier`. An average can move a lot because of a few extreme values. A median does not. So we use two simple checks. No new test is needed:

1. Compare the **medians** of the two groups.
2. Run the **same** t-test again **without** the flagged outliers.

Compare the medians

- **SPSS menus:** Analyze > Descriptive Statistics > Explore. Dependent List: price (not `ln_price`); Factor List: `host_type`; Display: Statistics (or Both); OK.
- **Syntax:**

* Compare the medians: price for each host type (read the Median rows of the Descriptives table).

```
EXAMINE VARIABLES=price BY host_type /PLOT=NONE /STATISTICS=DESCRIPTIVES.
```

In the **Descriptives** table, find the **Median** row for each host type. (If there is also a table for all listings together, use the one split by host type.)

Check: compare the two medians yourself. Are they far apart, or close together? Compare that with the gap between the two means in Part 1.

Run the test without the outliers

- **SPSS menus:** Data > Select Cases. Choose "If condition is satisfied", click If, type `price_outlier = 0`, Continue. Under Output choose "Filter out unselected cases"; OK. Then run the Part 1 t-test again (Analyze > Compare Means > Independent-Samples T Test).
- **Afterwards, switch the filter off:** Data > Select Cases > All cases > OK. If you forget, Parts 3, 4 and 6 silently use fewer listings.

- **Syntax:**

```
* Mark the listings that are not price outliers (1 = keep, 0 = set aside).  
COMPUTE keep_typical = (price_outlier = 0).
```

```
* Use only the kept listings in the next commands.  
FILTER BY keep_typical.
```

```
* Run the same t-test of price on the typical listings.  
T-TEST GROUPS=host_type(1 2) /VARIABLES=price.
```

```
* Switch the filter off so later parts use all listings again.  
FILTER OFF.
```

PSPP users

PSPP's Select Cases dialog is similar, but its wording differs: choose the condition option, enter `price_outlier = 0`, and make sure unselected cases are **filtered**, not deleted. Deleting removes the rows for the rest of the session. The syntax above (FILTER BY, then FILTER OFF) is the safest route.

Read the output and report it

Without the outliers, the Group Statistics table shows 6307 superhosts and 14185 other hosts. In the Welch row, Sig. (2-tailed) = .638.

With all listings, the difference looked significant ($p = .036$). Without the 745 Part 1 listings that are flagged as price outliers, the difference was not statistically significant (Welch's $t(11976.42) = -0.47$, $p = .638$, 95% CI [-6.21, 3.81]; Cohen's $d = -0.01$, tiny). A few extreme prices created the difference. Typical superhosts and other hosts charge about the same.

Lesson: Outliers can fool a t-test. The Part 1 result was significant only because of a few extreme prices. Always look at the data and the medians before you trust a p-value.

Stuck? Think of ten friends with similar incomes, and one billionaire joining the group. The average income jumps. The median hardly moves.

Part 3: Do guests rate cleanliness differently from location?

You are here: Part 3 of 5 (core).

Research question

Service-quality question: **For the same listing, do guests rate cleanliness differently from location?** If cleanliness scores are lower, hosts know where to improve.

Why a t-test fits

- The outcome is a **number**: a review score from 0 to 5.
- The two scores come from the **same listing**, so they are linked in pairs.
- We compare the **average difference** within each pair.

So we use a **paired t-test**.

Hypotheses

- **H0**: On average, cleanliness and location scores are the same.
- **H1**: On average, they are different.

Look first

Make one difference per listing (cleanliness minus location) and draw its histogram. Zero means "the same score".

- **SPSS menus**: Transform > Compute Variable. Target Variable: gap; Numeric Expression: review_scores_cleanliness - review_scores_location; OK. Then Graphs > Legacy Dialogs > Histogram; Variable: gap; OK.

Check: most listings get similar scores for both. Is the histogram centred a little below zero, at zero, or above zero?

Run the test

- **SPSS menus**: Analyze > Compare Means > Paired-Samples T Test. Move review_scores_cleanliness to Variable1 and review_scores_location to Variable2 (Pair 1); OK.
- **Syntax**:

```
* Make one difference per listing: cleanliness score minus location score.  
COMPUTE gap = review_scores_cleanliness - review_scores_location.
```

```
* Look first: histogram of the differences (0 means the same score).  
GRAPH /HISTOGRAM=gap.
```

```
* Run the paired t-test of cleanliness versus location.  
T-TEST PAIRS=review_scores_cleanliness WITH review_scores_location (PAIRED).
```

Read the output

The **Paired Samples Test** table shows the Mean of the paired differences (-.07), $t = -21.91$, $df = 21690$, and $\text{Sig. (2-tailed)} = .000$. Remember: ".000" means $p < .001$. PSPP's table rounds to two decimals (SPSS and Colab show more); the report sentence below uses three decimals, as the Colab notebook prints them.

How to report it

Across 21,691 listings, cleanliness scores were 0.066 points lower than location scores on average (paired t-test, $t(21690) = -21.91$, $p < .001$, 95% CI [-0.072, -0.060]). But the effect was tiny (Cohen's $d = -0.15$).

Lesson: With more than 21,000 listings, even a very small difference becomes “significant”. Significant does not mean important. Here $p < .001$, but Cohen's $d = -0.15$ is tiny. The effect size tells you whether it matters.

Stuck? A p-value answers "Is it real?". Cohen's d answers "Is it big?". You need both answers.

Part 4: Is the average listing below the superhost standard?

You are here: Part 4 of 5 (core).

Research question

Benchmarking question: superhosts are expected to keep an overall rating of at least 4.8.
Is the average NYC listing rated below 4.8?

Why a t-test fits

- The outcome is a **number**: the overall rating.
- We have **one group** (all rated listings) and **one fixed target** (4.8).

So we use a **one-sample t-test**.

Hypotheses

- **H0:** The average rating equals 4.8.
- **H1:** The average rating is different from 4.8.

Look first

- **SPSS menus:** Graphs > Legacy Dialogs > Histogram; Variable: review_scores_rating; OK.

Check: where is 4.8 on the histogram? Are most ratings above or below it?

Run the test

- **SPSS menus:** Analyze > Compare Means > One-Sample T Test. Test Variable(s): review_scores_rating; Test Value: 4.8; OK.
- **Syntax:**

* Look first: histogram of the overall ratings.
GRAPH /HISTOGRAM=review_scores_rating.

```
* Run the one-sample t-test of the overall rating against 4.8.  
T-TEST /TESTVAL=4.8 /VARIABLES=review_scores_rating.
```

Read the output

The **One-Sample Statistics** table shows $n = 21700$ and a mean of 4.73. The **One-Sample Test** table shows $t = -22.08$, $df = 21699$, $\text{Sig. (2-tailed)} = .000$ (so $p < .001$), and a Mean Difference of $-.07$ (the mean minus 4.8).

Two confidence intervals. SPSS and PSPP print the "95% Confidence Interval of the Difference", which is relative to 4.8 ($-.07$ to $-.06$). The report sentence uses the confidence interval of the **mean**, as the Colab notebook does: add 4.8 to the bounds to get the CI of the mean ($4.8 + (-.07) = 4.73$, and $4.8 + (-.06) = 4.74$).

How to report it

The average rating was 4.73 (SD = 0.45, $n = 21,700$; 95% CI of the mean 4.73 to 4.74), below the superhost standard of 4.8 (one-sample t-test, $t(21699) = -22.08$, $p < .001$; Cohen's $d = -0.15$, tiny).

Stuck? The confidence interval is the likely range for the true average. Here 4.8 is outside that range, so the average is probably different from 4.8.

Part 5: Your turn

You are here: Part 5 of 5 (core).

Question to explore: entire homes and private rooms. Which one costs more **per guest**? Use the listings that are not price outliers.

Before you run anything, write these in your notes:

1. **Research question:** _____
2. **Why a t-test fits:** Is the outcome a number? Are the groups separate or paired?
3. **Hypotheses:** H_0 : _____ H_1 : _____

Run the test

- **SPSS menus:** Transform > Compute Variable. Target Variable: price_per_guest; Numeric Expression: price / accommodates; OK. Data > Select Cases > If condition is satisfied: price_outlier = 0 AND room_type <= 2 (entire homes and private rooms, no outliers); Filter out unselected cases; OK. Analyze > Compare Means > Independent-Samples T Test. Test Variable: price_per_guest; Grouping Variable: room_type, groups 1 and 2; OK. Then switch the filter off (Data > Select Cases > All cases).
- **Syntax:**

```
* Make the nightly price per guest.  
COMPUTE price_per_guest = price / accommodates.
```

```

* Mark entire homes and private rooms that are not price outliers.
COMPUTE keep_home = (price_outlier = 0 AND room_type <= 2).

* Use only the marked listings in the next command.
FILTER BY keep_home.

* Run the t-test of price per guest for entire homes (1) versus private rooms
(2).
T-TEST GROUPS=room_type(1 2) /VARIABLES=price_per_guest.

* Switch the filter off so later parts use all listings again.
FILTER OFF.

```

Then read the Welch row, compute or read Cohen's d, and write the report sentence yourself.

Stuck? Copy the Part 1 report sentence and change the names and numbers. Remember: report the difference, the test, the p-value, the effect size, and the 95% CI.

Part 6: Going further (optional)

You are here: optional extra. Not graded. Skip it if you like.

Rank-based tests. These tests use ranks instead of averages. Use them when extreme values dominate. The **Mann-Whitney** test asks whether one group **tends to have higher values** than the other. The **Wilcoxon signed-rank** test asks whether the within-listing differences (cleanliness minus location) **tend to be above or below zero**.

- **Mann-Whitney:** two separate groups (like Part 1). SPSS menus: Analyze > Nonparametric Tests > Legacy Dialogs > 2 Independent Samples. Test Variable: price; Grouping Variable: host_type (1 and 2); Test Type: Mann-Whitney U; OK.
- **Wilcoxon:** paired scores (like Part 3). SPSS menus: Analyze > Nonparametric Tests > Legacy Dialogs > 2 Related Samples. Pair: review_scores_cleanliness and review_scores_location; Test Type: Wilcoxon; OK.

In both **Test Statistics** tables, read the row or column **Asymp. Sig. (2-tailed)**: it is the p-value.

Log price and Student's t-test. Taking the logarithm of price pulls extreme values in: run the Part 1 t-test on `ln_price`. Student's t-test assumes both groups have the same spread. Welch's does not, so Welch's is the safer default. Student's t-test is the "Equal variances assumed" row of the Part 1 output: compare its Sig. with the Welch row.

```

* Mann-Whitney test of price for superhosts versus other hosts (the Part 1 gr
oups).
NPAR TESTS /M-W=price BY host_type(1 2).

```

```
* Wilcoxon signed-rank test of cleanliness versus location (the Part 3 pairs).
NPAR TESTS /WILCOXON=review_scores_cleanliness WITH review_scores_location (P
AIRED).
```

```
* Welch t-test on log price for superhosts versus other hosts.
T-TEST GROUPS=host_type(1 2) /VARIABLES=ln_price.
```

Think about it: Does the Mann-Whitney test agree with Part 1's Welch test? Does the log-price t-test agree? Do the Student and Welch rows of Part 1 lead to the same conclusion at .05? What does that tell you about the Part 1 result?

Check your results

Compare your output with this table. The numbers are what PSPP 1.4.1 prints; SPSS may print more decimals, so round your numbers to the decimals in the table before you compare (and read **Two-Sided p** in SPSS 27 or later). For Parts 1 and 2, compare the "Equal variances not assumed" row.

	Part 1	Part 2	Part 3	Part 4
Test	Welch independent	Welch independent, outliers out	Paired	One-sample, test value 4.8
Groups or variables	Superhost / Other host	Superhost / Other host	Cleanliness / Location	Overall rating
N	6549 / 14688	6307 / 14185	21691	21700
Mean	267.63 / 281.81	215.89 / 217.09	4.68 / 4.74	4.73
Levene's F (Sig.)	6.45 (.011)	.02 (.879)	not used	not used
Mean Difference	-14.18	-1.20	-.07	-.07
t	-2.10	-.47	-21.91	-22.08
df	18708.20	11976.42	21690	21699
Sig. (2-tailed), as PSPP prints it	.036	.638	.000	.000
Report as	p = .036	p = .638	p < .001	p < .001
95% CI of the Difference	-27.43 to -.94	-6.21 to 3.81	-.07 to -.06	-.07 to -.06 (from 4.8)
Cohen's d (size)	-0.03 (tiny)	-0.01 (tiny)	-0.15 (tiny)	-0.15 (tiny)

Note. Parts 1 and 2 show the "Equal variances not assumed" (Welch) row. Cohen's d for Parts 1 and 2 is the SPSS-style d (mean difference / pooled SD); PSPP does not print it. The Colab notebook prints d = -0.029 for Part 1 and -0.007 for Part 2 (SPSS-style, to 3 decimals: -0.027 and -0.007); the size word is the same. For Parts 3 and 4 all tools use the same formula. See "Why your Cohen's d may differ slightly" in Part 1.

If your numbers differ: check that the filter is off (Data > Select Cases > All cases) before Parts 3 and 4; check that you defined the groups as 1 and 2; check that you tested price, not `ln_price`; and check that you read the Welch row.

Try it with AI

Copy this prompt into an AI tool, and paste your Part 1 output tables (Group Statistics and Independent Samples Test) after it:

"Here is SPSS output from an independent-samples t-test. Read the 'Equal variances not assumed' (Welch) row. Explain the result to a marketing manager in three short sentences. Mention the p-value and the effect size, and say whether the difference matters in practice."

Then check: did the AI read the Welch row? Did it mention the effect size? Did it say "significant" means "big"? (It should not.) If it reported a Sig. of ".000" as an exact p-value, correct it to " $p < .001$ ".

Data: Inside Airbnb (insideairbnb.com), NYC snapshot 2026-06-14, CC BY 4.0.